

Oussema Touhami

Computer Science Engineer – Specialization : New Technologies and Security

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Professional Profile

Software, Machine Learning and MLOps Engineer passionate about AI, ML, and building production-ready machine learning solutions. Experienced in developing ML pipelines, deploying models, and building scalable AI applications. Seeking a **Junior Software, Machine Learning, MLOps, or Data Science Engineer** opportunity.

Experience

- **Graduation Project (PFE) – Wevoo** February–June 2026

*Developed an end-to-end hybrid optimization framework combining heuristics, **Behavior Cloning**, and **PPO Reinforcement Learning** for warehouse optimization.*

- Built reproducible **MLOps** pipelines with **ZenML**, **MLflow**, **Docker**, and **CI/CD** for training and deployment.
- Awarded **18/20 (High Honors / Mention Très Bien)**; selected among the **best graduation projects** at ENSIT.

Technologies : PyTorch, OR-Tools, ZenML, MLflow, FastAPI, Streamlit, Docker, GitHub Actions.

- **Artificial Intelligence Internship – Insight Plus** June–August 2025

Development of a multilingual chatbot (French / Tunisian) for currency exchange operations.

Technologies : **Prompt Engineering**, **LangChain**, **LLM**, **RAG (ChromaDB, OllamaEmbeddings)**, **FastAPI**, **n8n**, **Angular**, **MongoDB**.

- **Web Development Internship – Sagemcom** July–August 2024

Application for managing electronic board test files.

Technologies : **Angular**, **Flask**, **MongoDB**.

Projects

• **NLP Project : Fine-Tuning DistilBERT for Emotion Classification**

Training a Transformer model (DistilBERT) on the `dair-ai/emotion` dataset.

Technologies : **Python**, **PyTorch**, **Transformers (HuggingFace)**, **Datasets**, **Matplotlib**.

• **MLOps Project : Intelligent Customer Ticket Classification**

*Complete MLOps pipeline using two NLP approaches : **Transformer (Hugging Face)** and **TF-IDF & SVM** with **CI/CD**, **monitoring**, and an **AI agent** orchestrating predictions.*

Technologies : **FastAPI**, **MLflow**, **DVC**, **Docker**, **GitHub Actions**, **Prometheus**, **Grafana**, **scikit-learn**.

• **AI Project : Medical RAG Chatbot**

*A fast, accurate, and private medical information chatbot powered by **Retrieval-Augmented Generation (RAG)** using **medical PDFs**.*

Technologies : **Python**, **Streamlit**, **FAISS**, **LLM API**, **HuggingFace Sentence Transformers**, **Langchain**.

• **DevOps Project** : Implementation of a full **CI/CD** pipeline for a **Springboot** application with **Jenkins**, integrating **SonarQube** for code analysis and **ArgoCD** for **GitOps**-based deployment on **Kubernetes**.

• **Medical Web Application with Diabetes Prediction (PFA2)**

2025 – ENSIT

*Development of a diabetes prediction system using **SVM**, **Random Forest**, and **Decision Tree** models.*

Technologies : **Angular**, **SpringBoot**, **Flask**, **PostgreSQL**.

• **Medical Management Mobile Application**

Technologies : **Ionic**, **Angular**, **Flask**, **MongoDB**, **SQLite**.

Education

– **Computer Engineering Degree** — Higher National School of Engineering of Tunis (ENSIT) 2023–2026

– **Prep Cycle in Physics–Chemistry** — Prep Institute for Engineering Studies of Tunis (IPEIT) 2021–2023

– **Baccalaureate in Experimental Sciences (Honors : Very Good)** — Esprit Private High School 2021

Technical Skills

- **AI** : Machine Learning, LangChain, NLP, RAG, Prompt Engineering, ChromaDB, n8n, Scikit-Learn, Hugging-Face
- **Frontend** : Angular, JavaFX, HTML/CSS, Streamlit
- **DevOps** : Git, Docker, GitHub Actions, GitLab, Kubernetes, Helm, Terraform, ArgoCD, SonarQube, Jenkins
- **Backend** : SpringBoot, Flask, FastAPI, ExpressJS, JEE
- **Databases** : MySQL, PostgreSQL, MongoDB
- **Languages** : C/C++, Python, JavaScript, TypeScript, Java

Volunteering : Treasurer – IEEE ENSIT Student Branch 2024–2025

Award : **Outstanding Student Branch of the Year (Large Category)**.

Organization and participation in multiple IEEE events.